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Optimal Electrode Positions for an SSVEP-based BCI

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***Abstract*—In this paper, the performance of a steady-state visually evoked potential (SSVEP)-based brain-computer interface (BCI) was evaluated after reducing the number of electroencephalography (EEG) electrodes used for recording. Our aim was the comparison of the performance using a different number of electrodes, as well as determining the best electrode locations. The main benefit of determining the most efficient electrode sites can be the shortening of the preparation time needed for the setup of the BCI system. The performance of the BCI was measured with a different number of EEG electrodes reduced from initially sixteen to six electrodes as the intermediate step, and finally to four electrodes. Seventeen subjects participated in this study, seven were male and ten female. The mean ITRs were 27.50 bit/min, 24.09 bit/min, and 23.23 bit/min for sixteen, six and four EEG electrodes, respectively. However, not all users managed to use the system with less electrodes, as one user did not manage to control the system with six electrodes, and four of the seventeen users were not able to finish the spelling tasks with four electrodes. With a reduced number of electrodes, a high accuracy can be achieved while retaining a fair spelling speed.**

***Index Terms*—Brain-Computer Interface (BCI), Steady-State Visually Evoked Potential (SSVEP), Electrode Number Reduction, Electroencephalography (EEG), Human-Machine Interface**

I. INTRODUCTION

Brain-computer interfaces (BCIs) allow communication using only brain signals, so that muscular movements become dispensable. This makes them particularly well suited for communication for people with motor disabilities, who are unable to express themselves otherwise.

BCIs can use various methods for generating brain signals. They can be endogenous, using techniques that do not require any external stimulation, such as imaginary movement [1], or exogenous, using something to stimulate the brain in order to produce a specific natural neural activity [2], [3]. In this study, an exogenous BCI based on SSVEP was used. These signals are responses to visual stimuli flickering at constant frequencies. Following this stimulation, the same frequency (and its harmonics) can be detected from the brain signals derived from the visual cortex of the BCI user.

In the modern BCIs, the brain signals are usually recorded non-invasively, using EEG. EEG requires a good connection between the electrodes and the scalp to provide quality recordings. Usually, non-abrasive electrode gel is applied during the preparation for EEG recording. The drawbacks of the gel-based system are the time it takes to apply the gel correctly and the uncomfortableness of the gel itself, which needs to be washed out of the hair after the EEG recording.

A high number of electrodes is often used in BCI applications in order to achieve the best accuracies due to the higher amount of data available for processing. With a reduced number of electrodes, a good connection between the scalp and the electrodes is essential. Each electrode placed on the head of a user designates a certain EEG channel. By removing the electrodes reaching the lowest weights in contributing to the signal, a faster preparation of the participant can be achieved, while comparatively high accuracies can still be reached during the completion of spelling tasks. It also reduces the cost of hardware, less data has to be improved by filters, and in the case of wireless acquisition less data needs to be sent and received, so the computational power and the signal bandwidth requirements can be lowered. The signal bandwidth can be problematic when using wireless electrodes, since the amount of data that can be transferred is limited in size.

The average classification accuracy for gel electrodes is usually higher than that of dry-contact electrodes and water-based electrodes [4]. Gel electrodes reached in that cited study mean classification accuracies of 96%, whereas dry and water-based electrodes reached 63% and 88%, respectively. Gel electrodes can be differentiated as active and passive electrodes. In contrast to passive electrodes, active electrodes are not as affected by cable movements and other external noises, due to the pre-amplification of the measured electrical activity close to the recording sites. Impedances below 5kΩ are recommended to achieve a good signal quality for passive electrodes [5].

An example of a BCI speller is the multi-phase SSVEP speller, also discussed in [6], where a letter is written in two or three steps. For both, the two- and three-step speller, the canonical correlation analysis (CCA) was used with eight electrodes.

Many opinions and results state that a different number of EEG channels is needed, depending on the testing method. One study showed that the minimum number of electrodes needed for multi-target SSVEP-BCI control is 3 electrodes when an accuracy above 70% is to be achieved [7]. The best channel combinations include P_Z, O_Z and O_1, and optimal channel sets always include P_Z. This study used two different classification methods, the already mentioned CCA and minimum energy combination method (MEC), to explore the minimum number of signal electrodes [7]. The best channel configuration for both testing methods was determined offline. This study showed that MEC benefits from more channels used, but in total, no significant difference was found between the signal processing methods.

Another study investigated age-related differences in SSVEP-based BCI performance [8], where the mean accuracy and ITR using eight electrodes were 98.49% and 27.36 bit/min, and 91.13% and 16.10 bit/min for the young and old group, respectively. A three-step speller interface was used in this study [8].

The study that aimed at improving the sequential floating forward selection method (SFFS) for channel selection [9] used 59 channels and utilised, besides the improved SFFS, the support vector machine recursive feature elimination (SVM-RFE) method. It found that the improved method reduced the computation time and focused on the entire channel configuration rather than an individual channel. The SVM-RFE method was slower, but focused on the individual channel configuration, while removed channels were not revisited.

Another study found that when testing with nine and five electrodes, the data length was important to achieve high ITRs. The highest accuracies (89.83%) and ITR (325.33 bit/min) were reached with nine electrodes [10].

In this study, a three-step speller was used, which requires three selections to be made to select a letter similarly as in [8]. Three-step spellers are very robust, they only require four different stimulation frequencies, they are well-tested and reliable, which is needed for testing the differences in the performance depending on external factors, such as the number of used EEG electrodes.

This study is different from the presented studies above, since an online test was conducted and the evaluation of the best working electrodes was done on site, allowing the observation of the performance of the participants with a reduced number of EEG electrodes. The reduction of electrodes was done by keeping the best working six electrodes after the first run, which was done with all sixteen electrodes. The same process was done to get the four best working electrodes. Compared to [9], the goal was not to reduce the computation time, but to find the minimum number of channels and electrode locations working best for most people, while reaching a high accuracy.

If only four electrodes are required, the overall process will be much faster and more convenient, and therefore closer to being applicable in everyday life.

II. Materials and Methods

A. Participants

Seventeen healthy volunteers (ten females) with an average age of 22.94 years ($\pm$SD 3.19) participated in this study. All participants gave written informed consent before the experiment, in accordance with the Declaration of Helsinki. This study was approved by the ethical committee of the medical faculty of the University Duisburg-Essen. All information was stored anonymously during the experiment and results cannot be traced back to the participants. Subjects had the opportunity to opt-out of the study at any time. Spectacles were worn when appropriate. The subjects received a small financial reward for participation in this study.

B. Speller Interface

The interface consisted of four boxes arranged in a 2x2 matrix, flickering at set frequencies of 6.32 Hz, 7.06 Hz, 8.00 Hz and 9.23 Hz (Fig. 1). Three selections were required to write a letter, similarly as in [8]. In the initial layout, the upper two boxes and the lower left box contained letter groups from the English alphabet “A-I”, “J-R” and “S-Z”, with the lower left box also containing the space command “_”. The fourth box corresponded to the “Delete” or “Back” command, depending on the steps the user made towards writing a letter. Starting from the initial layout, the content of the boxes was changed when a selection was made. The first three boxes would contain three letter long groups with the letters of the previously selected box (e.g. “A-C”, “D-F”, “G-I” if the box “A-I” was selected). A further selection would result in individual letters in the boxes, while selecting an individual letter resulted in writing it.

The command “Delete” could only be selected in the initial layout, and it deleted the last written character, whereas “Back” could be seen in the second and third layouts and was associated with the command to go back to the previous layout. Users had to type the target words accurately; all mistakes had to be corrected with the help of these commands. Ideally, three steps were needed to spell a letter, but any misclassification would increase the number of steps and the time needed, while also decreasing the accuracy.

The interface contained the target word and the text already written by the user. After each classification, the subjects received audio feedback assisting them, so they did not need to check after every step whether progress has been made or misspellings happened.

C. Signal Acquisition

Subjects were seated in front of an LCD screen (BenQ XL24IIT, resolution: 1920×1080 pixels, vertical refresh rate: 120 Hz) at a distance of about 80 cm and were instructed to avoid unnecessary movements, since these affect the quality of the recordings negatively. The computer operated with Microsoft Windows 7 Enterprise running on an Intel processor ”Intel Core i7-3770”. Ag/AgCl electrodes (EASYCAP GmbH, Herrsching, Germany) were used to acquire the EEG signals. These were easily removable from the cap during recordings, which was important for the reduction of number of electrodes during the online experiment. Standard abrasive electrolytic electrode gel was applied between the electrodes and the scalp. Usually, impedances below 5 kΩ are needed for a good quality recording, but to ensure high accuracy even after removing electrodes, all impedances were lowered below 3 kΩ. A g.USBamp (Guger Technologies, Graz, Austria) EEG amplifier was utilized for the recording.

The sampling frequency was set to 256 Hz. A digital band pass filter (between 2 and 100 Hz) and a notch filter (around 50 Hz) were applied during signal acquisition into the amplifier directly. The data were sent from the amplifier in blocks of 64 samples.

The reference electrode was located at C_Z, and the ground electrode at AF_Z. For the experiment, sixteen electrodes were used: P_Z, P_3, P_4, P_5, PO_3, POO_1, POO_2, PO_4, P_6, PO_7, O_1, O_Z, O_2, PO_8, O_9, and O_{10}, according to the international 10/5 system for electrode placement. The experimental setup after the completed first phase of the experiment is shown in the Fig. 1 and Fig. 2. During the experiment, the users

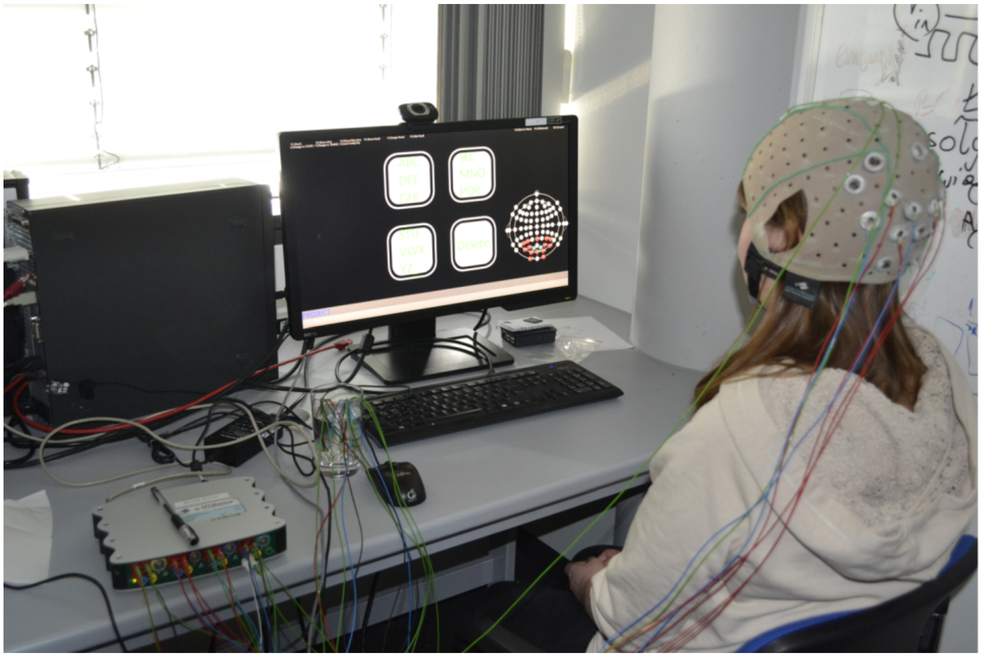

Fig. 1. Setup of the experiment in which the participant is connected with 6 electrodes after the first spelling phase. The three-step spelling interface with the spellig task “PROJECT” in the bottom row is shown on the monitor. Removed electrodes are in a glass of water, connected to the reference electrode to cancel out these electrode signals. The schematic next to the boxes shows the calculated weights for the specific electrodes online color coded, based on the calculated values.

had to perform six online spelling tasks, where the number of electrodes was decreased gradually by removing them from the EEG cap and connecting them to the reference. The number of electrodes was reduced at first to the best six, then to the best four electrodes. The selection of the best electrodes and further details is described in the section Electrode Selection Procedure.

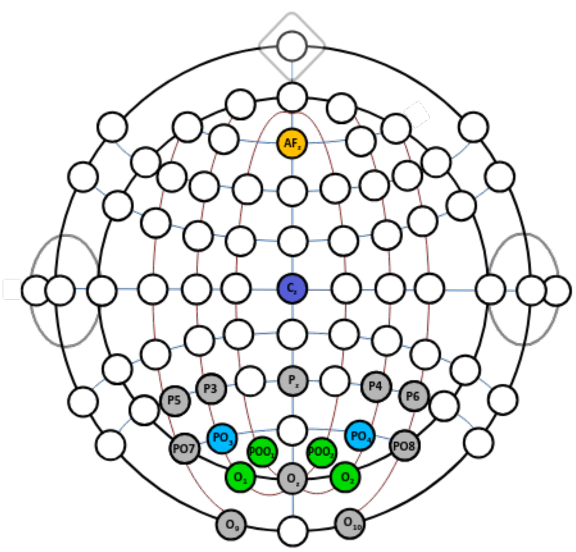

Fig. 2. Schematic, showing the positions of the four best working electrodes in green, and the next two best electrodes in blue. Yellow marks the used ground electrode, while C_Z, marked purple shows the used reference electrode location.

D. Signal Processing

For processing the recorded brain activity, minimum energy combination (MEC) was utilized [11]. The EEG data were processed online by several classifiers, which differed only in the amount of data processed. Six of them were running concurrently, as long as enough data were available: 1 s, 1.5 s, 2 s, 4 s, 6 s and 8 s of data were processed simultaneously. The probability of the resulting output class needed to surpass a pre-set threshold (individual to each stimulus frequency), and if it was not reached, the output was rejected. In that case, a new block of data from the amplifier shuffled out the oldest block of data and another calculation began with the newer EEG data. For more details please see [12]. The electrodes used in this study were combined during the classification process to achieve the least amount of noise [13]. As the classification combines the EEG recordings into arbitrary channels, the calculation of SNR (signal to noise ratio) will not influence the classification as much as this selection process. With other classification methods however, SNR could be a good indicator of electrode usefulness.

The thresholds were set manually at the start of the experiment, before the first spelling tasks. The principle was to allow fast selections while being as close to 100% accuracy as possible. The thresholds were kept the same throughout the experiment, regardless of the number of electrodes.

After every classification, the flickering and data processing stopped for two seconds, to allow users to find the next target. Without this break the lingering effects of the previously attended stimulus would bias the next classification results, however two seconds are enough for this effect to disappear from the neural activity. The break in between the classifications was kept the same during the whole experiment. The value of the break combined with the minimum amount of data the classifiers processed limited the maximum achievable ITR of this study to 40.00 bit/min.

E. Experimental Setup

After signing the consent form, the participants had to fill out a pre-questionnaire. The age of the participants ranged from 19-33 and five users already used a BCI before. Shortly after, the participants were prepared for the experiment by putting on an EEG cap and lowering the impedances. By asking users to focus on the flickering boxes, the thresholds were set individually for each box. In case the participant struggled to select a certain box during this check, the corresponding threshold was lowered. The frequencies used for each box are described in the section Speller Interface.

To avoid bias from not knowing how to operate the BCI, the subjects were then instructed to spell a short word to familiarize with the interface. After the familiarization run, the participants were asked to perform two spelling tasks, writing the words "PROJECT" and "TRAINS". Between the tasks, users were allowed to rest. In order to avoid bias from user tiredness the duration of the experiment was kept under one hour, therefore the number of spelling tasks was limited to six (two with each electrode setup).

The target words were chosen by taking the number of selections for each of the three boxes into account. When combining both words, these are almost equal. In total, the first, second and third box would have been chosen 14, 13 and 12 times, respectively, during a perfect run. If a wrong selection was made, the subjects had to correct this by selecting the fourth box ("Back"/"Delete"). When the target word was spelled correctly, the spelling phase ended. If the user was not able to finish a spelling task, due to too many wrong inputs or if it was not possible to exceed the critical value required to make a selection for an extended duration, the task was stopped manually by the experimenter.

Following the first two spelling tasks (first phase), the experimenter checked the measurements and selected the six best working electrodes by reviewing the electrode recommendations, as described in the next section. For phase two (Fig. 1), the same two words had to be written by the participants, which was followed by the removal of two further electrodes for phase three. The experiment ended after the participants wrote both target words with four electrodes. If spelling in phase two or three was very inaccurate or if the completion of a phase was not progressing after several minutes, the experiment was stopped. These tasks are referred to as not finished in the Results section.

F. Electrode Selection Procedure

MEC combines the data from the different electrodes by assigning weights to every one of them based on the amount of noise (any non-useful information for the BCI) in the recorded data. The level of noise is determined by removing the components the stimuli can cause (the same frequency as the stimulus as well as its harmonics) from the recorded data and analyzing the rest of the data, to determine which electrodes had the highest amount of noise. The noisiest electrodes are excluded from further calculations, which provide the classifier output.

Throughout both spelling tasks, the normalized weights assigned to each electrode during classification were saved whenever a target was selected. The average weight of each electrode was used to select the best electrodes (the highest average weight). For the second and third phase of the experiment, the six and four best were used, respectively.

The reduction of electrode numbers was executed by removing the electrodes with low average weights from the EEG cap and putting them in water, which was also connected to the reference electrode socket of the amplifier. This essentially made these electrodes measure the same values as the reference channel, which resulted in close to zero values in the acquired data. This setup allowed a quick and simple removal of electrodes without the need of altering the settings of the software and guaranteed the correctness of the performed procedure, and without the possibility of causing any issues in the software.

III. Results

The mean time it took the participants to finish the spelling tasks with the different electrode configurations increased with the decreasing number of channels. The mean spelling times were 87.73 s, 114.60 s and 105.90 s for sixteen, six and four electrodes used, respectively. Users who were not able to finish all spelling phases (subject 8, 12, 13 and 17) were not included in this calculation. Some participants took longer to finish the spelling task for either one of the target words or both. Some participants were not able to or had difficulties using the speller with six electrodes or less, so the wrong selections were also an indication that they would not be able to use the system with four electrodes. The accuracies and ITRs were calculated by using the formulas given in [2] and were measured regarding the spelled letters, not the EEG directly.

The performance of the system with all subjects is shown in Table I. As expected with this reliable speller, all participants were able to spell the target words with sixteen electrodes. Excluding the subjects that were not able to complete all phases, the mean accuracy for the first spelling task was 94.61% (±3.69%) with a mean ITR of 27.50 bit/min (±4.73 bit/min). With a decreasing number of electrodes, the number of participants that were not able to spell the words increased. For six electrodes, the participants achieved a mean accuracy and mean ITR of 91.27% (±8.06%) and 24.09 bit/min (±8.73 bit/min), respectively. When further reduced to four electrodes, the mean accuracy increased to 93.22% (±5.63%), while the mean ITR value decreased to 23.23 bit/min (±7.10 bit/min).

The lowest ITR stayed above 10 bit/min (out of the maximum 40 bit/min). The lowest accuracies were reached by subject 7, spelling the first target word with six electrodes and subject 8, spelling the second target word with 6 electrodes, reaching 67.7% and 63.8%, respectively. The minimum accuracy for the spelling task with sixteen electrodes was 72.2% and and 71% with four electrodes. The shortest spelling times were achieved when all sixteen electrodes were used. Participants that used

TABLE I
The averaged accuracy [%], ITR [bit/min] and spelling time [s] for the different electrode configurations are represented in this table. The sign "-" represents the inability of the user to complete the spelling phase with a certain number of electrodes used. The mean values and standard deviations are given at the bottom of the table for each column. The values not used in the calculations are marked with "*".

	Accuracy [%]			ITR [bit/min]			Spelling Time [s]		
Subject	16 electrodes	6 electrodes	4 electrodes	16 electrodes	6 electrodes	4 electrodes	16 electrodes	6 electrodes	4 electrodes
1	95.35	80.40	87.50	24.70	13.65	19.25	86.40	154.50	121.50
2	97.85	94.85	83.40	31.10	29.15	18.88	73.95	80.65	119.70
3	94.45	100.00	95.85	27.25	33.00	24.00	87.20	70.80	98.10
4	95.35	94.45	100.00	29.00	28.75	30.40	75.40	79.30	77.40
5	90.30	98.00	100.00	28.85	35.25	32.70	78.45	67.55	71.45
6	91.95	95.35	88.85	24.80	22.00	16.80	83.80	97.35	125.50
7	93.10	72.10	91.85	29.40	10.46	20.05	77.55	200.00	107.00
8	100.00*	81.90*	-	33.90*	11.73*	-	69.30*	311.50*	-
9	97.50	90.00	97.85	35.45	27.35	32.85	63.05	83.70	68.95
10	86.10	95.45	97.50	22.20	27.10	28.20	137.70	80.80	79.20
11	97.85	95.85	97.85	27.90	20.30	20.35	82.05	114.50	111.20
12	100.00*	88.35*	-	30.65*	10.24*	-	76.30*	272.00*	-
13	87.40*	-	-	12.85*	-	-	155.50*	-	-
14	100.00	83.55	87.80	25.90	12.65	14.55	90.15	250.50	150.00
15	98.00	100.00	97.50	34.20	38.20	32.55	69.80	61.20	68.70
16	92.15	86.50	85.85	16.75	15.30	11.45	135.00	149.50	177.50
17	94.45*	94.85*	-	29.35*	29.30*	-	79.20*	81.70*	-
Mean	**94.61**	**91.27**	**93.22**	**27.50**	**24.09**	**23.23**	**87.73**	**114.60**	**105.90**
SD	**3.85**	**8.39**	**5.86**	**4.92**	**9.09**	**7.39**	**22.82**	**57.99**	**33.47**

the BCI speller before did not score higher in accuracy or ITR or needed less time to finish the spelling task compared to other users. One of the participants, experienced in using a BCI, was not able to operate the system with reduced number of electrodes.

Performing two single factor ANOVA tests on the ITR and accuracy of the spelling tasks "TRAINS" and "PROJECT" combined, all three electrode configurations were compared to each other. The p-values for the accuracy (0.41) and ITR (0.30) both indicate that there is no statistically significant difference between the performance with sixteen, six and four electrodes. However these results do not take into account the fact that 23.53% (four out of seventeen) of our subjects could not use the system with only four electrodes. This has to be taken into account when designing experiments with a minimal number of electrodes.

Throughout the experiment the combined weight of the four best electrodes working for 82.4% of the participants (14 out of 17) reached a mean of 88.83% of the total assigned weights, meaning that even when sixteen electrodes were used, these four would provide more than 88.8% of the data used for the classification. The highest combined weight was 96.89% of subject 4, while the lowest 2 values were 75.51% and 79.65% of subjects 9 and 16, respectively.

Table II shows the channels chosen for the second and third phases (six and four electrodes). During the test, POO_1, POO_2, O_2 and O_1 were the best working electrodes, which means that they were the most often chosen for the phases two and three for the majority of the subjects, as shown in Fig. 2, indicated by green dots. POO_1 and POO_2 were the most often used electrodes for the six electrode and four electrode setup. For the channel POO_1, the frequency of being selected was 87.5% and 84.62% for six and four electrodes, respectively. POO_2 was chosen 100% for the second spelling phase and 69.23% when four electrodes were selected. O_2 appeared in a total of 81.25% (six electrodes) and 53.85% (four electrodes), whereas O_1 appeared by 75% (six electrodes) and 53.85% (four electrodes). PO_3 (62.5% and 38.46% for the second and third spelling phase) and PO_4 (62.5% and 23.08% for the six and four electrode configuration) were the second best working electrodes, represented in blue in Fig. 2. This was evaluated by choosing the best working electrode after testing with sixteen, six and four electrodes, respectively. When looking at Table II, it can be seen that when using four electrodes, only three participants did not have the channel POO_1 included in their selection of the four best electrodes.

IV. Discussion & Conclusion

For some of the subjects, a much better accuracy was reached when tested with four electrodes compared to six electrodes.

TABLE II
The table shows the selected electrode locations when using six and four electrodes. The electrodes kept after the first electrode reduction are listed and the chosen channels kept after the second phase are marked with bold letters.

Subject	Selected EEG channels					
1	$\mathbf{PO_3}$	$\mathbf{POO_1}$	$\mathbf{POO_2}$	PO_4	$\mathbf{O_1}$	O_2
2	PO_3	$\mathbf{POO_1}$	$\mathbf{POO_2}$	PO_4	$\mathbf{O_1}$	$\mathbf{O_2}$
3	$\mathbf{PO_3}$	$\mathbf{POO_1}$	$\mathbf{POO_2}$	PO_7	$\mathbf{O_1}$	O_2
4	P_3	P_5	$\mathbf{POO_1}$	$\mathbf{POO_2}$	$\mathbf{O_Z}$	$\mathbf{O_2}$
5	POO_1	$\mathbf{POO_2}$	$\mathbf{PO_4}$	O_1	$\mathbf{O_2}$	$\mathbf{PO_8}$
6	P_3	$\mathbf{PO_3}$	$\mathbf{POO_1}$	POO_2	$\mathbf{O_1}$	$\mathbf{O_Z}$
7	$\mathbf{PO_3}$	$\mathbf{POO_1}$	POO_2	$\mathbf{PO_7}$	$\mathbf{O_1}$	O_2
8	POO_2	PO_4	PO_7	O_1	O_2	O_10
9	PO_3	$\mathbf{POO_1}$	$\mathbf{POO_2}$	PO_4	$\mathbf{O_1}$	$\mathbf{O_2}$
10	PO_3	$\mathbf{POO_1}$	$\mathbf{POO_2}$	$\mathbf{PO_4}$	O_Z	$\mathbf{O_2}$
11	PO_3	POO_1	$\mathbf{POO_2}$	$\mathbf{PO_4}$	$\mathbf{O_2}$	$\mathbf{PO_8}$
12	POO_2	PO_4	PO_7	O_1	O_2	PO_8
13	-	-	-	-	-	-
14	PO_3	$\mathbf{POO_1}$	POO_2	$\mathbf{PO_7}$	$\mathbf{O_1}$	$\mathbf{O_Z}$
15	$\mathbf{POO_1}$	$\mathbf{POO_2}$	PO_4	O_1	$\mathbf{O_Z}$	$\mathbf{O_2}$
16	$\mathbf{P_3}$	$\mathbf{P_5}$	$\mathbf{PO_3}$	$\mathbf{POO_1}$	POO_2	PO_4
17	POO_1	POO_2	O_1	O_Z	O_2	PO_8

To further investigate this effect, additional experiments need to be conducted, with more participants. Our null hypothesis (that there is no difference between the means) could not be rejected, which suggests that there is no significant decline in performance when using less electrodes.

As shown in Table I, the time it took the subjects to spell the target words, just as the amount of overall selections, increased only slightly with less electrodes. The mean accuracy shows a slight decreasing tendency but it is not statistically significant. Likewise, the ITR shows a non-significant slightly decreasing tendency.

The four best electrode positions were not the same for each subject. The four electrodes selected indicated in bold letters in the chart included three of the four best working electrodes combined with either PO_3, PO_4 or O_z. PO_7 was one of the best four electrodes twice, while PO_8 and P_3 appeared only once. For more details, please see Table II. The best electrode positions were located over the primary visual cortex on the back side of the head. The evaluation of the results suggests these four electrodes for use in SSVEP-based BCIs. If the four electrodes are not the best combination, they can still be considered as a good and reliable selection. To make the results more accurate, electrodes that scored as second best working electrodes could be added.

In this experiment, it was tested if the reduction of electrodes would still provide a reliable BCI, resulting in a high accuracy and ITR and maintaining a fair spelling time. According to the presented results, reducing the number of electrodes from sixteen to four yields no issues for 76.47% of subjects; however, the rest of the subjects could not use the system with four electrodes. Even in those cases, electrode number reduction is still possible, to some intermediate value, like six electrodes.

In conclusion, it can be said that with a reduced number of electrodes, the accuracy and ITR of the utilized BCI application are not affected in most cases. When the best working electrodes are found for the individual, a high accuracy can be achieved while maintaining a high spelling speed. This means that electrode preparation times can be reduced, eliminating some of the uncomfortableness caused by the electrolytic gel, while still resulting in a fast and reliable BCI system. In future experiments, additional tests can be done with even less electrodes, or finding an automatic way of determining the best working electrodes. We also suggest tests with longer tasks that result in more robust results, as well as finding the minimal number of electrodes that works with every single user. It is also planned to investigate why some subjects were able to use the speller with six electrodes, reaching an accuracy higher than 80%, but were unable to operate the system with four electrodes. Moreover, the effect on the subjects mentally when schemes like the selected electrodes are shown on the screen next to the flickering boxes (Fig. 1) needs to be investigated in further experiments.

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